

SU2 Represenations And Iso-Spin Matrices

Baijayanta Bhattacharyya

Feb 2025

Contents

List Of Figures	iii
1 Isospin Matrics for 2 Level system	1
1.1 2 Level System	1
1.1.1 Computation of Commutators	2
1.1.2 Solving for a	2
1.1.3 Matrix Representation	3
2 Isospin Matrics for 3 Level system	5
2.1 3 Level System	5
2.1.1 Computation of Commutators	6
2.1.2 Solving for a, b	6
2.1.3 Matrix Representation	7
3 Isospin Matrics for 4 Level system	9
3.1 4 Level System	9
3.1.1 Computation of Commutators	10
3.1.2 Solving for a, b, c	10
3.1.3 Matrix Representation	11
4 For N Level System	14

List of Figures

1.1	2 Level System	1
2.1	Caption	5
3.1	Caption	9

To my parents,

Thank you for your endless love, guidance, and unwavering support.

Everything I am is because of you.

CHAPTER 1

Isospin Matrices for 2 Level system

1.1 2 Level System

We have total 2 levels, labeled by $|+\rangle, |-\rangle$. Where the action of operator τ_+ takes us one step up in the ladder and τ_- takes us one step down. There are no states when τ_+ acting on $|+\rangle$ and τ_- acting on $|-\rangle$. This is shown schematically in Figure (1.1). So these Ladder Operators τ_+ and τ_- can be written

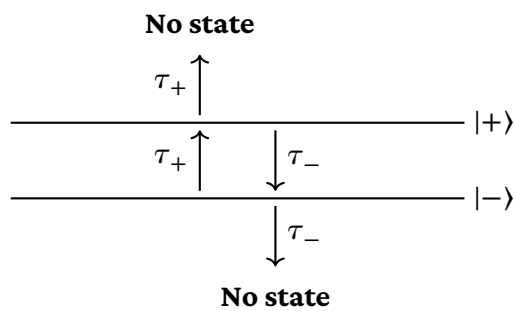


Figure 1.1: 2 Level System

this way:

$$\tau_+ = a |+\rangle\langle -| \quad (1.1)$$

$$\tau_- = a^* |-\rangle\langle +| \quad (1.2)$$

1.1.1 Computation of Commutators

$$\begin{aligned}
 [\tau_+, \tau_-] &= [a|+\rangle\langle-|, a^*|-\rangle\langle+|] \\
 &= aa^*|+\rangle\langle+| - aa^*|-\rangle\langle-| \\
 &= aa^* \left[|+\rangle\langle+| - |-\rangle\langle-| \right] \\
 &= 2\tau_3
 \end{aligned}$$

So we have:

$$\tau_3 = \frac{aa^*}{2} \left[|+\rangle\langle+| - |-\rangle\langle-| \right] \quad (1.3)$$

Now using this commutator identity we find out values of a .

$$[\tau_3, \tau_+] = \tau_+ \quad (1.4)$$

$$\begin{aligned}
 [\tau_3, \tau_+] &= \frac{aa^*}{2} \left[|+\rangle\langle+| - |-\rangle\langle-|, a|+\rangle\langle-| \right] \\
 &= \frac{aa^*}{2} \left[a|+\rangle\langle-| + a|+\rangle\langle-| \right] \\
 &= aa^* \left[a|+\rangle\langle-| \right] \\
 &= a|+\rangle\langle-| \\
 &= \tau_+
 \end{aligned}$$

Comparing, we get:

$$aa^* = 1 \quad (1.5)$$

1.1.2 Solving for a

Considering a to be real we get:

$$a = 1 \quad (1.6)$$

Thus,

$$\tau_+ = |+\rangle\langle-| \quad (1.7)$$

$$\tau_- = |-\rangle\langle+| \quad (1.8)$$

1.1.3 Matrix Representation

The matrix Representation of τ_+ and τ_- is given by:

$$\tau_+ = \begin{pmatrix} \langle +|\tau_+|+ \rangle & \langle +|\tau_+|- \rangle \\ \langle -|\tau_+|+ \rangle & \langle -|\tau_+|- \rangle \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

and,

$$\tau_- = \begin{pmatrix} \langle +|\tau_-|+ \rangle & \langle +|\tau_-|- \rangle \\ \langle -|\tau_-|+ \rangle & \langle -|\tau_-|- \rangle \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

and, We define

$$\tau_{\pm} = \tau_x \pm i\tau_y \quad (1.9)$$

So by Eq. (1.9) we get :

$$\tau_x = \frac{1}{2}(\tau_+ + \tau_-)$$

$$\tau_x = \frac{1}{2} \left[\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \right] = \frac{1}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

and,

$$\tau_y = \frac{1}{2i}(\tau_+ - \tau_-)$$

$$\tau_y = \frac{1}{2i} \left[\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \right] = \frac{1}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

Finally:

$$\tau_3 = \tau_z = \tau_3 = \frac{aa^*}{2} \left[|+\rangle\langle +| - |-\rangle\langle -| \right] = \frac{1}{2} \left[|+\rangle\langle +| - |-\rangle\langle -| \right]$$

$$\tau_3 = \begin{pmatrix} \langle +|\tau_3|+ \rangle & \langle +|\tau_3|- \rangle \\ \langle -|\tau_3|+ \rangle & \langle -|\tau_3|- \rangle \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

And we can find out $\tau^2 = \tau_x^2 + \tau_y^2 + \tau_z^2 = \frac{3}{4} \mathbb{1}_{2 \times 2} = \frac{1}{2} \left(1 + \frac{1}{2} \right) \mathbb{1}_{2 \times 2}$. For this we have $\tau = \frac{1}{2}$. This are the iso-spin matrices and the represents a doublet(2 states). As $[\tau^2, \tau_3] = 0$ they can have simultaneous eigen-functions and these states are generally labelled by $|\tau, \tau_3\rangle$. We get the states:

$$1. |+\rangle = |\tau = \frac{1}{2}, \tau_3 = +\frac{1}{2}\rangle = \left| \frac{1}{2}, \frac{1}{2} \right\rangle$$

$$2. |-\rangle = |\tau = \frac{1}{2}, \tau_3 = -\frac{1}{2}\rangle = \left| \frac{1}{2}, -\frac{1}{2} \right\rangle$$

They differ by iso-spin 3rd component. And those states are given by

$$|+\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, |-\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad (1.10)$$

For example the baryons p, n have the same iso-spin $\tau = \frac{1}{2}$, but differ only in 3rd component $(p)_z = \frac{1}{2}, (n)_z = -\frac{1}{2}$. They are the iso-spin doublets.

Similarly K^+, K^0 are the isospin doublets, with $\tau = \frac{1}{2}, (K^+)_z = \frac{1}{2}, (K^0)_z = -\frac{1}{2}$.

CHAPTER 2

Isospin Matrices for 3 Level system

2.1 3 Level System

Consider we have three energy levels, labelled by $|+\rangle$, $|0\rangle$, $|-\rangle$. Where the action of operator τ_+ takes us one step up in the ladder and τ_- takes us one step down. There are no states when τ_+ acting on $|+\rangle$ and τ_- acting on $|-\rangle$. This is shown schematically in Figure (2.1). So these Ladder Operators τ_+ and τ_- can be written this way:

$$\tau_+ = a |0\rangle\langle -| + b |+\rangle\langle 0| \quad (2.1a)$$

$$\tau_- = a^* |-\rangle\langle 0| + b^* |0\rangle\langle +| \quad (2.1b)$$

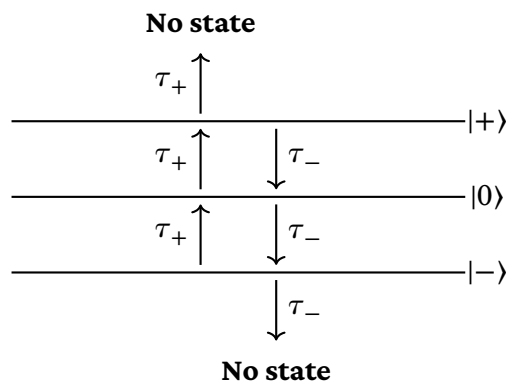


Figure 2.1: Caption

2.1.1 Computation of Commutators

$$\begin{aligned}
[\tau_+, \tau_-] &= [a|0\rangle\langle-| + b|+\rangle\langle 0|, a^*|-\rangle\langle 0| + b^*|0\rangle\langle+|] \\
&= aa^*|0\rangle\langle 0| + bb^*|+\rangle\langle+| - aa^*|-\rangle\langle-| - bb^*|0\rangle\langle 0| \\
&= (aa^* - bb^*)|0\rangle\langle 0| + bb^*|+\rangle\langle+| - aa^*|-\rangle\langle-| \\
&= 2\tau_3
\end{aligned}$$

So we have:

$$\tau_3 = \frac{1}{2} \left[(aa^* - bb^*)|0\rangle\langle 0| + bb^*|+\rangle\langle+| - aa^*|-\rangle\langle-| \right] \quad (2.2)$$

Now using this commutator identity we find out values of a, b .

$$[\tau_3, \tau_+] = \tau_+ \quad (2.3)$$

$$\begin{aligned}
[\tau_3, \tau_+] &= \frac{1}{2} \left[(aa^* - bb^*)|0\rangle\langle 0| + bb^*|+\rangle\langle+| - aa^*|-\rangle\langle-|, a|0\rangle\langle-| + b|+\rangle\langle 0| \right] \\
&= \frac{1}{2} \left[(aa^* - bb^*) a|0\rangle\langle-| + b^2 b^*|+\rangle\langle 0| + a^2 a^*|0\rangle\langle-| - (aa^* - bb^*) b|+\rangle\langle 0| \right] \\
&= \frac{1}{2} \left[(2aa^* - bb^*) a|0\rangle\langle-| - (aa^* - 2bb^*) b|+\rangle\langle 0| \right] \\
&= \frac{2aa^* - bb^*}{2} a|0\rangle\langle-| - \frac{aa^* - 2bb^*}{2} b|+\rangle\langle 0| \\
&= a|0\rangle\langle-| + b|+\rangle\langle 0| \\
&= \tau_+
\end{aligned}$$

Comparing, we get:

$$2aa^* - bb^* = 2 \quad (2.4)$$

$$aa^* - 2bb^* = -2 \quad (2.5)$$

2.1.2 Solving for a, b

From Eq. (2.4) and Eq. (2.5) we get:

$$\begin{aligned}
2aa^* - bb^* &= 2 = -aa^* + 2bb^* \\
\Rightarrow aa^* &= bb^* = 2
\end{aligned}$$

Considering a, b to be real we get:

$$a = b = \sqrt{2} \quad (2.6)$$

Thus,

$$\tau_+ = \sqrt{2} |0\rangle\langle -| + \sqrt{2} |+\rangle\langle 0| \quad (2.7a)$$

$$\tau_- = \sqrt{2} |-\rangle\langle 0| + \sqrt{2} |0\rangle\langle +| \quad (2.7b)$$

2.1.3 Matrix Representation

The matrix Representation of τ_+ and τ_- is given by:

$$\tau_+ = \begin{pmatrix} 0 & b & 0 \\ 0 & 0 & a \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & \sqrt{2} & 0 \\ 0 & 0 & \sqrt{2} \\ 0 & 0 & 0 \end{pmatrix}$$

and,

$$\tau_- = \begin{pmatrix} 0 & 0 & 0 \\ b^* & 0 & 0 \\ 0 & a^* & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ \sqrt{2} & 0 & 0 \\ 0 & \sqrt{2} & 0 \end{pmatrix}$$

We define

$$\tau_{\pm} = \tau_x \pm i\tau_y \quad (2.8)$$

So by Eq. (2.8) we get :

$$\tau_x = \frac{1}{2}(\tau_+ + \tau_-)$$

$$\tau_x = \frac{1}{2} \begin{pmatrix} 0 & b & 0 \\ b^* & 0 & 0 \\ 0 & a^* & 0 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 0 & \sqrt{2} & 0 \\ \sqrt{2} & 0 & \sqrt{2} \\ 0 & \sqrt{2} & 0 \end{pmatrix} = \begin{pmatrix} 0 & \frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & \frac{1}{\sqrt{2}} & 0 \end{pmatrix}$$

and,

$$\tau_y = \frac{1}{2i}(\tau_+ - \tau_-)$$

$$\tau_y = \frac{1}{2i} \begin{pmatrix} 0 & b & 0 \\ -b^* & 0 & 0 \\ 0 & -a^* & 0 \end{pmatrix} = \frac{1}{2i} \begin{pmatrix} 0 & \sqrt{2} & 0 \\ -\sqrt{2} & 0 & \sqrt{2} \\ 0 & -\sqrt{2} & 0 \end{pmatrix} = \begin{pmatrix} 0 & -\frac{i}{\sqrt{2}} & 0 \\ \frac{i}{\sqrt{2}} & 0 & -\frac{i}{\sqrt{2}} \\ 0 & \frac{i}{\sqrt{2}} & 0 \end{pmatrix}$$

Finally:

$$\tau_3 = \tau_z = \frac{1}{2} \left[(aa^* - bb^*)|0\rangle\langle 0| + bb^*|+\rangle\langle +| - aa^*|-\rangle\langle -| \right] = \frac{1}{2} \left[bb^*|+\rangle\langle +| - aa^*|-\rangle\langle -| \right]$$

$$\tau_3 = \frac{1}{2} \begin{pmatrix} bb^* & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -aa^* \end{pmatrix} = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -2 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

And we can find out $\tau^2 = \tau_x^2 + \tau_y^2 + \tau_z^2 = 2\mathbb{I}_{3 \times 3} = 1(1+1)\mathbb{I}_{3 \times 3}$. For this we have $\tau = 1$. This are the iso-spin matrices and the represents a triplet(3 states). As $[\tau^2, \tau_3] = 0$ they can have simultaneous eigen-functions and these states are generally labelled by $|\tau, \tau_3\rangle$. We get the states:

1. $|+\rangle = |\tau = 1, \tau_3 = 1\rangle = |1, 1\rangle$
2. $|0\rangle = |\tau = 1, \tau_3 = 0\rangle = |1, 0\rangle$
3. $|-\rangle = |\tau = 1, \tau_3 = -1\rangle = |1, -1\rangle$

They differ by iso-spin 3rd component. And those states are given by

$$|+\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, |0\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, |-\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \quad (2.9)$$

For example the baryons $\Sigma^+, \Sigma^0, \Sigma^-$ have the same iso-spin $\tau = 1$, but differ only in 3rd component $(\Sigma^+)_z = 1, (\Sigma^0)_z = 0, (\Sigma^-)_z = -1$. They are the iso-spin triplets. Where a convention has been used that most +ve one has given the highest isospin 3rd component.

CHAPTER 3

Isospin Matrices for 4 Level system

3.1 4 Level System

We have total 4 levels, labeled by $|1\rangle, |2\rangle, |3\rangle, |4\rangle$. Where the action of operator τ_+ takes us one step up in the ladder and τ_- takes us one step down. There are no states when τ_+ acting on $|4\rangle$ and τ_- acting on $|1\rangle$. This is shown schematically in Figure (3.1). So these Ladder Operators τ_+ and τ_- can be written this way:

$$\tau_+ = a |2\rangle\langle 1| + b |3\rangle\langle 2| + c |4\rangle\langle 3| \quad (3.1)$$

$$\tau_- = a^* |1\rangle\langle 2| + b^* |2\rangle\langle 3| + c^* |3\rangle\langle 4| \quad (3.2)$$

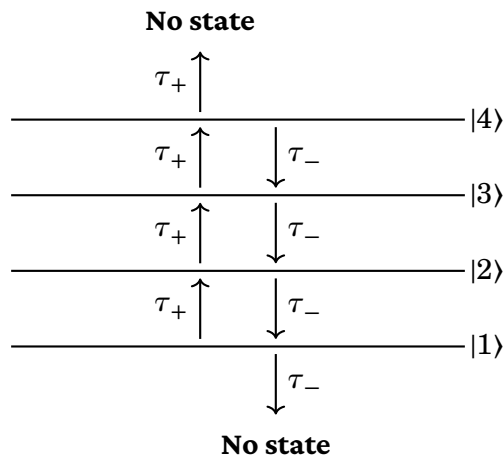


Figure 3.1: Caption

3.1.1 Computation of Commutators

$$\begin{aligned}
[\tau_+, \tau_-] &= [a|2\rangle\langle 1| + b|3\rangle\langle 2| + c|4\rangle\langle 3|, a^*|1\rangle\langle 2| + b^*|2\rangle\langle 3| + c^*|3\rangle\langle 4|] \\
&= aa^*|2\rangle\langle 2| + bb^*|3\rangle\langle 3| + cc^*|4\rangle\langle 4| - aa^*|1\rangle\langle 1| - bb^*|2\rangle\langle 2| - cc^*|3\rangle\langle 3| \\
&= (aa^* - bb^*)|2\rangle\langle 2| + (bb^* - cc^*)|3\rangle\langle 3| + cc^*|4\rangle\langle 4| - aa^*|1\rangle\langle 1| \\
&= 2\tau_3
\end{aligned}$$

So we have:

$$\tau_3 = \frac{1}{2} \left[(aa^* - bb^*)|2\rangle\langle 1| + (bb^* - cc^*)|3\rangle\langle 3| + cc^*|4\rangle\langle 4| - aa^*|1\rangle\langle 1| \right] \quad (3.3)$$

Now using this commutator identity we find out values of a, b, c .

$$[\tau_3, \tau_+] = \tau_+ \quad (3.4)$$

$$\begin{aligned}
[\tau_3, \tau_+] &= \frac{1}{2} \left[(aa^* - bb^*)|2\rangle\langle 2| + (bb^* - cc^*)|3\rangle\langle 3| + cc^*|4\rangle\langle 4| - aa^*|1\rangle\langle 1|, a|2\rangle\langle 1| + b|3\rangle\langle 2| + c|4\rangle\langle 3| \right] \\
&= \frac{1}{2} \left[(aa^* - bb^*) a|2\rangle\langle 1| + (bb^* - cc^*) b|3\rangle\langle 2| + c^2 c^*|4\rangle\langle 3| \right. \\
&\quad \left. + a^2 a^*|2\rangle\langle 1| (aa^* - bb^*) b|3\rangle\langle 2| - (bb^* - cc^*) c|4\rangle\langle 3| \right] \\
&= \frac{aa^* - bb^* + aa^*}{2} a|2\rangle\langle 1| + \frac{2bb^* - aa^* - cc^*}{2} b|3\rangle\langle 2| + \frac{cc^* - bb^* + cc^*}{2} c|4\rangle\langle 3| \\
&= a|2\rangle\langle 1| + b|3\rangle\langle 2| + c|4\rangle\langle 3| \\
&= \tau_+
\end{aligned}$$

Comparing, we get:

$$2aa^* - bb^* = 2 \quad (3.5)$$

$$2bb^* - cc^* - aa^* = 2 \quad (3.6)$$

$$2cc^* - bb^* = 2 \quad (3.7)$$

3.1.2 Solving for a, b, c

Adding Eq. (3.5) and Eq. (3.7) we get :

$$2aa^* + 2cc^* = 2 + 2bb^*$$

$$aa^* + cc^* = 2 + bb^*$$

From Eq. (3.6) we get;

$$2bb^* - (aa^* + cc^*) = 2$$

$$2bb^* - 2 - bb^* = 2$$

$$bb^* = 4$$

We take a, b, c to be real:

$$b = \sqrt{4} = 2$$

From Eq. (3.5) and Eq. (3.7) we get:

$$2a^2 = 2c^2 = 2 + b^2 = 2 + 4 = 6$$

$$a = c = \sqrt{3} \quad (3.8)$$

Thus,

$$\tau_+ = \sqrt{3} |2\rangle\langle 1| + 2 |3\rangle\langle 2| + \sqrt{3} |4\rangle\langle 3| \quad (3.9)$$

$$\tau_- = \sqrt{3} |1\rangle\langle 2| + 2 |2\rangle\langle 3| + \sqrt{3} |3\rangle\langle 4| \quad (3.10)$$

3.1.3 Matrix Representation

The matrix Representation of τ_+ and τ_- is given by:

$$\tau_+ = \begin{pmatrix} 0 & c & 0 & 0 \\ 0 & 0 & b & 0 \\ 0 & 0 & 0 & a \\ 0 & 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & \sqrt{3} & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & \sqrt{3} \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

and,

$$\tau_- = \begin{pmatrix} 0 & 0 & 0 & 0 \\ c^* & 0 & 0 & 0 \\ 0 & b^* & 0 & 0 \\ 0 & 0 & a^* & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ \sqrt{3} & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & \sqrt{3} & 0 \end{pmatrix}$$

We define

$$\tau_{\pm} = \tau_x \pm i\tau_y \quad (3.11)$$

So by Eq. (3.11) we get :

$$\tau_x = \frac{1}{2}(\tau_+ + \tau_-)$$

$$\tau_x = \frac{1}{2} \begin{pmatrix} 0 & c & 0 & 0 \\ c^* & 0 & b & 0 \\ 0 & b^* & 0 & a \\ 0 & 0 & a^* & 0 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 0 & \sqrt{3} & 0 & 0 \\ \sqrt{3} & 0 & 2 & 0 \\ 0 & 2 & 0 & \sqrt{3} \\ 0 & 0 & \sqrt{3} & 0 \end{pmatrix} = \begin{pmatrix} 0 & \frac{\sqrt{3}}{2} & 0 & 0 \\ \frac{\sqrt{3}}{2} & 0 & 1 & 0 \\ 0 & 1 & 0 & \frac{\sqrt{3}}{2} \\ 0 & 0 & \frac{\sqrt{3}}{2} & 0 \end{pmatrix}$$

and,

$$\tau_y = \frac{1}{2i}(\tau_+ - \tau_-)$$

$$\tau_y = \frac{1}{2i} \begin{pmatrix} 0 & c & 0 & 0 \\ -c^* & 0 & b & 0 \\ 0 & -b^* & 0 & a \\ 0 & 0 & -a^* & 0 \end{pmatrix} = \frac{1}{2i} \begin{pmatrix} 0 & \sqrt{3} & 0 & 0 \\ -\sqrt{3} & 0 & 2 & 0 \\ 0 & -2 & 0 & \sqrt{3} \\ 0 & 0 & -\sqrt{3} & 0 \end{pmatrix} = \begin{pmatrix} 0 & -\frac{i\sqrt{3}}{2} & 0 & 0 \\ \frac{i\sqrt{3}}{2} & 0 & -i & 0 \\ 0 & -i & 0 & -\frac{i\sqrt{3}}{2} \\ 0 & 0 & \frac{i\sqrt{3}}{2} & 0 \end{pmatrix}$$

Finally:

$$\tau_3 = \tau_z = \frac{1}{2} [(aa^* - bb^*) |2\rangle\langle 2| + (bb^* - cc^*) |3\rangle\langle 3| + cc^* |4\rangle\langle 4| - aa^* |1\rangle\langle 1|]$$

$$\begin{aligned} \tau_3 &= \frac{1}{2} \begin{pmatrix} cc^* & 0 & 0 & 0 \\ 0 & bb^* - cc^* & 0 & 0 \\ 0 & 0 & aa^* - bb^* & 0 \\ 0 & 0 & 0 & -aa^* \end{pmatrix} \\ &= \begin{pmatrix} \frac{3}{2} & 0 & 0 & 0 \\ 0 & \frac{4-3}{2} & 0 & 0 \\ 0 & 0 & \frac{3-4}{2} & 0 \\ 0 & 0 & 0 & -\frac{3}{2} \end{pmatrix} \\ &= \begin{pmatrix} \frac{3}{2} & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & -\frac{1}{2} & 0 \\ 0 & 0 & 0 & -\frac{3}{2} \end{pmatrix} \end{aligned}$$

And we can find out $\tau^2 = \tau_x^2 + \tau_y^2 + \tau_z^2 = \frac{15}{4} \mathbb{1}_{4 \times 4} = \frac{3}{2} \left(\frac{3}{2} + 1 \right) \mathbb{1}_{4 \times 4}$. For this we have $\tau = \frac{3}{2}$. This are the iso-spin matrices and the represents a quartet(4 states). As $[\tau^2, \tau_3] = 0$ they can have simultaneous eigen-functions and these states are generally labelled by $|\tau, \tau_3\rangle$. We get the states:

1. $|4\rangle = |\tau = 3/2, \tau_3 = 3/2\rangle = |\frac{3}{2}, \frac{3}{2}\rangle$
2. $|3\rangle = |\tau = 3/2, \tau_3 = 1/2\rangle = |\frac{3}{2}, \frac{1}{2}\rangle$
3. $|2\rangle = |\tau = 3/2, \tau_3 = -1/2\rangle = |\frac{3}{2}, -\frac{1}{2}\rangle$
4. $|1\rangle = |\tau = 3/2, \tau_3 = -3/2\rangle = |\frac{3}{2}, -\frac{3}{2}\rangle$

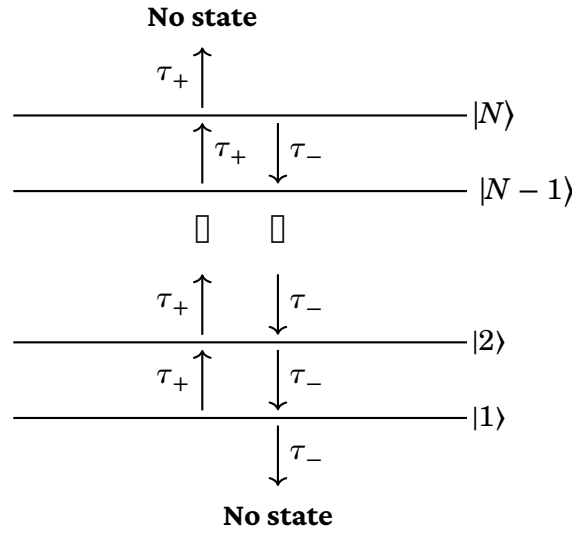
They differ by iso-spin 3rd component. And those states are given by

$$|4\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, |3\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, |2\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, |1\rangle = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \quad (3.12)$$

For example the baryons $\Delta^{++}, \Delta^+, \Delta^0, \Delta^-$ have the same iso-spin $\tau = \frac{3}{2}$, but differ only in 3rd component $(\Delta^{++})_z = \frac{3}{2}, (\Delta^+)_z = \frac{1}{2}, (\Delta^0)_z = -\frac{1}{2}, (\Delta^-)_z = -\frac{3}{2}$. They are the iso-spin quartets.

CHAPTER 4

For N Level System



As we have seen for 2 level system we have $\tau = \frac{1}{2}$, 3 level system we have $\tau = 1$, 4 level system we have $\tau = \frac{3}{2}$. In general given the value of τ it represents total $2\tau + 1$ no of states. To represent N states we must have $\tau = \frac{N-1}{2}$. As N is always a positive integer, this forces τ to be either an integer or half integer. The iso-spin operators have the matrix form,

$$\tau_+ = \begin{pmatrix} 0 & b_\tau & 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & b_{\tau-1} & 0 & \cdots & 0 & 0 \\ 0 & 0 & 0 & b_{\tau-2} & \cdots & 0 & 0 \\ 0 & 0 & 0 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & 0 & b_{-\tau+1} \\ 0 & 0 & 0 & 0 & \cdots & 0 & 0 \end{pmatrix}$$

$$\begin{aligned}
\tau_- &= \begin{pmatrix} 0 & 0 & 0 & 0 & \cdots & 0 & 0 \\ b_\tau & 0 & 0 & 0 & \cdots & 0 & 0 \\ 0 & b_{\tau-1} & 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & b_{\tau-2} & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 0 & 0 & \cdots & b_{-\tau+1} & 0 \end{pmatrix} \\
\tau_x &= \frac{1}{2} \begin{pmatrix} 0 & b_\tau & 0 & 0 & \cdots & 0 & 0 \\ b_\tau & 0 & b_{\tau-1} & 0 & \cdots & 0 & 0 \\ 0 & b_{\tau-1} & 0 & b_{\tau-2} & \cdots & 0 & 0 \\ 0 & 0 & b_{\tau-2} & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & 0 & b_{-\tau+1} \\ 0 & 0 & 0 & 0 & \cdots & b_{-\tau+1} & 0 \end{pmatrix} \quad (4.1) \\
\tau_y &= \frac{1}{2} \begin{pmatrix} 0 & -ib_\tau & 0 & 0 & \cdots & 0 & 0 \\ ib_\tau & 0 & -ib_{\tau-1} & 0 & \cdots & 0 & 0 \\ 0 & ib_{\tau-1} & 0 & -ib_{\tau-2} & \cdots & 0 & 0 \\ 0 & 0 & ib_{\tau-2} & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & 0 & -ib_{-\tau+1} \\ 0 & 0 & 0 & 0 & \cdots & ib_{-\tau+1} & 0 \end{pmatrix} \quad (4.2) \\
\tau_z &= \begin{pmatrix} \tau & 0 & 0 & \cdots & 0 \\ 0 & \tau-1 & 0 & \cdots & 0 \\ 0 & 0 & \tau-2 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & -\tau \end{pmatrix} \quad (4.3)
\end{aligned}$$

where

$$b_j \equiv \sqrt{(\tau+j)(\tau+1-j)}.$$